The setup last time

V = vector space with basis $\mathcal{B} = \{b_1, \dots, b_n\}$

$T: V \to V$ is a linear transformation.

$$\begin{tikzcd} v \arrow[r] \arrow[d] & T(v) \arrow[d] \\ {[v]_{\mathcal{B}}} \arrow[r, "{[T]_{\mathcal{B}}}"'] & {[T(v)]_{\mathcal{B}}} \end{tikzcd}$$

Here, $[T]_{\mathcal{B}} \in M_{n \times n}$ is given by

$$[T]_{\mathcal{B}} = \left[[T(b_1)]_{\mathcal{B}} \cdots [T(b_n)]_{\mathcal{B}} \right], \text{ and}$$

$$[T(v)]_{\mathcal{B}} = [T]_{\mathcal{B}} [v]_{\mathcal{B}}. \quad (1)$$

Q How does this setup change when we choose a different basis $\mathcal{A} = \{a_1, \dots, a_n\}$ for V?

A We still have $[T(v)]_{\mathcal{A}} = [T]_{\mathcal{A}} [v]_{\mathcal{A}}$ (2)

From Thm 15 in §4.6 we have

$$[v]_{\mathcal{A}} = \underset{\mathcal{A} \leftarrow \mathcal{B}}{P} [v]_{\mathcal{B}}, \quad \underset{\mathcal{A} \leftarrow \mathcal{B}}{P} = \left[[b_1]_{\mathcal{A}} \cdots [b_n]_{\mathcal{A}} \right]$$

$$[T(v)]_{\mathcal{A}} = \underset{\mathcal{A} \leftarrow \mathcal{B}}{P} [T(v)]_{\mathcal{B}}$$

Substituting into ②, we obtain

$$\underset{A\leftarrow B}{P}\,[T(v)]_B = [T]_A \underset{A\leftarrow B}{P}\,[v]_B \quad ③$$

Recall that $\underset{A\leftarrow B}{P}$ is invertible with $\left(\underset{A\leftarrow B}{P}\right)^{-1} = \underset{B\leftarrow A}{P}$

Multiplying ③ by $\left(\underset{A\leftarrow B}{P}\right)^{-1}$, we obtain

$$[T(v)]_B = \left(\underset{A\leftarrow B}{P}\right)^{-1} [T]_A \underset{A\leftarrow B}{P}\,[v]_B. \quad ④$$

Comparing ④ with ①, we see that

$$[T]_B = \left(\underset{A\leftarrow B}{P}\right)^{-1} [T]_A \underset{A\leftarrow B}{P}$$

In particular the two matrices $[T]_B$ and $[T]_A$ are similar. By Thm 4 §5.2 $[T]_B$ and $[T]_A$ have the same characteristic polynomial. Therefore they have the same eigenvalues. This tells us the definition of the characteristic polynomial of T is well-defined (does not depend on choice of basis.)

Appendix B <u>Complex numbers</u>

Definitions

$i = \sqrt{-1}$ or simply $i^2 = -1$.

$\mathbb{C}$ = complex numbers

$= \{ a + bi : a, b \in \mathbb{R} \}$

standard form

$(a+bi) + (c+di) = (a+c) + (b+d)i$

$(a+bi)(c+di) = ac + adi + bci + bdi^2$

$= (ac - bd) + (ad + bc)i$

Nice rules from $\mathbb{R}$ about addition and multiplication are also true for $\mathbb{C}$.

$\mathrm{Re}(a+bi) = a$, $\mathrm{Im}(a+bi) = b$.

Let $z = a + bi \in \mathbb{C}$

z is <u>imaginary</u> if $a = 0$, i.e. $z = bi$ or $\mathrm{Re}(z) = 0$.

z is <u>real</u> if $b = 0$ i.e. $z = a$ or $\mathrm{Im}(z) = 0$.

The <u>complex conjugate</u> of z is

$\bar{z} = a - bi$

$z\bar{z} = (a+bi)(a-bi) = a^2 - \cancel{abi} + \cancel{abi} - b^2 i^2$

$= a^2 + b^2 \geq 0$ (and real)

Thm

Let $w, z \in \mathbb{C}$. Then

1. $\overline{w+z} = \bar{w} + \bar{z}$

2. $\overline{wz} = \bar{w}\,\bar{z}$

3. $z\bar{z} \geq 0$

4. $\bar{\bar{z}} = z$

Def. Let $z = a+bi \in \mathbb{C}$. The <u>absolute value</u> (<u>modulus</u>) of z is

$$|z| = \sqrt{z\bar{z}} = \sqrt{a^2+b^2} \geq 0$$

$$|1+3i| = \sqrt{1^2+3^2} = \sqrt{10}.$$

Theorem Let $w, z \in \mathbb{C}$.

Then
1. $z\bar{z} = |z|^2$
2. $|wz| = |w|\,|z|$
3. $|w+z| \leq |w| + |z|$ (Triangle inequality)

Observe that if $z = a+bi \neq 0$ then $a \neq 0$ or $b \neq 0$.

$\iff a^2 > 0$ or $b^2 > 0$

$\iff a^2+b^2 > 0.$

$\iff z\bar{z} > 0$

$\iff |z| > 0.$

This proves $z = 0 \iff |z| = 0$.

In addition if $z \neq 0$ then $|z| \neq 0$ and

$$z\left(\bar{z}/|z|^2\right) = \frac{z\bar{z}}{|z|^2} = \frac{|z|^2}{|z|^2} = 1$$

This proves $\frac{1}{z} = z^{-1} = \frac{\bar{z}}{|z|^2}$.

For example $(1-2i)^{-1} = \frac{1+2i}{\sqrt{5}} = \frac{1}{\sqrt{5}} + \frac{2}{\sqrt{5}} i$

Remarks. The two most important facts about the complex numbers are

1. $\mathbb{C}$ behaves in the same way as $\mathbb{R}$ under addition and multiplication.

2. Every polynomial p with coefficients in $\mathbb{C}$ can be written as $p(x) = z(x-z_1)^{n_1} \cdots (x-z_k)^{n_k}$ where $z, z_1, \ldots, z_k \in \mathbb{C}$. (The Fundamental Theorem of Algebra).

From now on we can consider vector spaces over $\mathbb{C}$.

For example $\mathbb{C}^2 = \left\{ \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} : z_1, z_2 \in \mathbb{C} \right\}$

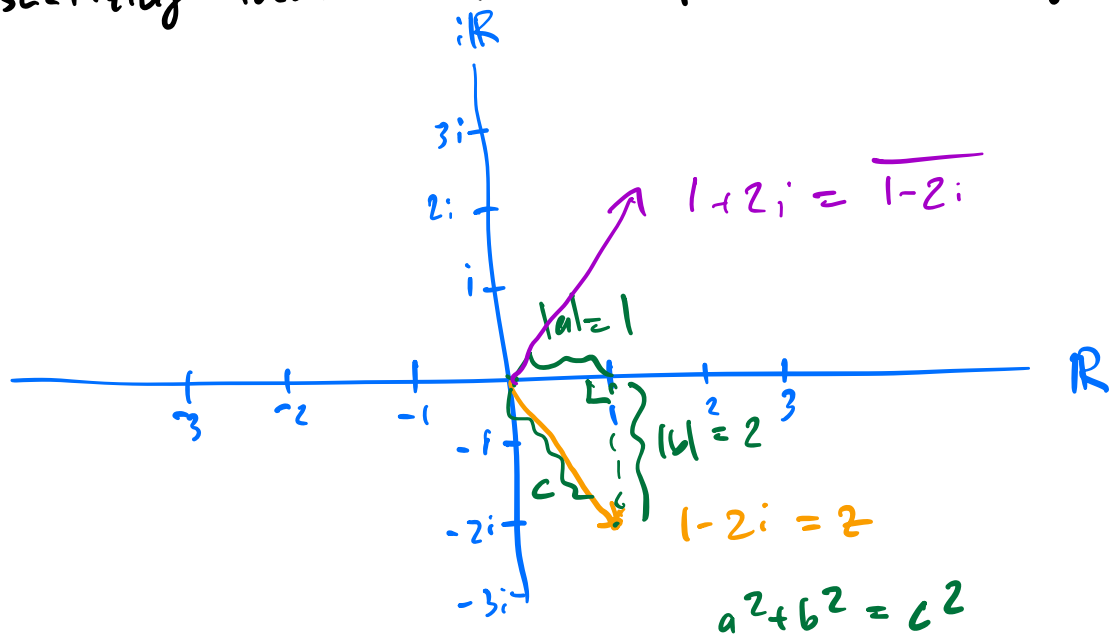
$$= \left\{ \begin{bmatrix} a+bi \\ c+di \end{bmatrix} : a, b, c, d \in \mathbb{R} \right\}$$

or $M_{2\times 2}(\mathbb{C}) = \left\{ \begin{pmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{pmatrix} : z_{11}, z_{12}, z_{21}, z_{22} \in \mathbb{C} \right\}$

Because of ② every linear transformation $T: V \to V$ of a <u>complex</u> vector space has eigenvalues.

Back to complex numbers ...

We can interpret complex numbers geometrically by visualizing them in the complex plane (Argand plane)

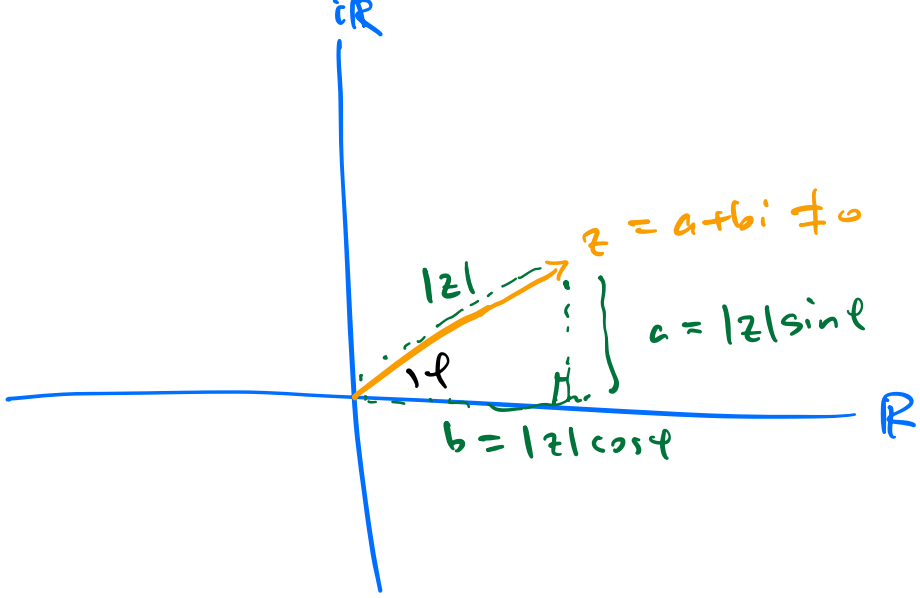


$a^2 + b^2 = c^2$

$\Rightarrow z\bar{z} = c^2 \Rightarrow c = \sqrt{z\bar{z}} = |z|$

$|z|$ = length of "vector" z

Visualizing multiplication is more delicate. For this we put complex numbers into polar form



φ is the angle from the +ve real axis to the vector representation $z \in \mathbb{C}$, in the counterclockwise direction. Using trigonometry, we see from the sketch above that

$$\begin{aligned} z &= a+bi \\ &= |z|\cos\varphi + (|z|\sin\varphi)i \\ &= |z|(\cos\varphi + i\sin\varphi) \end{aligned}$$

The short form for $\cos\varphi + i\sin\varphi$ is $e^{i\varphi}$.

i.e. $e^{i\varphi} = \cos\varphi + i\sin\varphi$ by <u>definition</u>

The angle φ is called the <u>argument</u> of z.

$z = |z|e^{i\varphi}$ is called the <u>polar form</u> of $z \neq 0$.

If we restrict the argument (angle) to be

$0 \leq \varphi < 2\pi$ then we obtain a

<u>unique</u> expression of z in polar form.